

CyberShield AI

Month 1 Project Planning & Research Report

1. Goal

The primary goal of the first month of the **CyberShield AI** project is to understand the project requirements, conduct detailed research on modern cyber security threats, and create a well-structured development plan. This phase focuses on identifying the major security challenges faced by individuals and organizations, studying common cyber attacks, and selecting appropriate technologies for building the platform.

During this stage, research is carried out on various cyber security concepts such as phishing attacks, SQL injection, cross-site scripting (XSS), malware, password security, and network attacks. Based on this research, the project's objectives, system architecture, technology stack, database design, and development roadmap are planned.

Completing this planning phase ensures that the project has a strong foundation before implementation begins. It also helps reduce development risks, improves project organization, and provides a clear direction for future development phases.

2. Project Title

CyberShield AI – AI-Powered Cyber Security Awareness, Detection & Attack Simulation Platform

Description:

CyberShield AI is a web-based cyber security platform designed to educate users about cyber threats while providing practical tools for detecting and analyzing security risks. The platform combines artificial intelligence with cyber security concepts to help students, professionals, and organizations understand modern cyber attacks in a safe and controlled environment.

The project includes multiple features such as website security scanning, email phishing detection, password strength analysis, malware file scanning, network scanning, attack simulations (SQL Injection, XSS, CSRF, MITM, DDoS, and Phishing), security awareness training, AI-powered assistance, threat intelligence, and report generation.

The main objective of CyberShield AI is to improve cyber security awareness, provide hands-on learning through safe simulations, and assist users in identifying potential security threats before they become serious risks. The platform is designed for educational purposes and can also be used by organizations to enhance cyber security knowledge and best practices.

3. Problem Statement

Cyber attacks are increasing rapidly across the world, affecting individuals, businesses, educational institutions, and government organizations. Many users are unaware of common cyber threats such as phishing attacks, malicious websites, weak passwords, malware infections, and social engineering techniques. Due to a lack of cyber security awareness, users often become victims of data theft, financial fraud, identity theft, and unauthorized access to their personal or organizational information.

There is a growing need for an educational platform that not only explains these cyber threats but also allows users to experience them in a safe and controlled environment. Traditional learning methods are often theoretical and do not provide practical understanding of how attacks occur or how they can be prevented.

CyberShield AI addresses this problem by providing an AI-powered cyber security awareness, detection, and attack simulation platform. It offers interactive tools such as website security scanning, phishing email analysis, password strength evaluation, malware scanning, network scanning, attack simulations, and an AI security assistant. These features help users understand cyber threats, learn secure practices, and improve their overall cyber security knowledge without exposing them to real-world risks.

4. Research Topics

During the first month of the CyberShield AI project, extensive research was conducted on various cyber security concepts and attack techniques to understand current cyber threats and identify the features required for the platform. The following topics were studied:

4.1 Phishing Attacks

Phishing is a cyber attack in which attackers use fake emails, websites, or messages to trick users into revealing sensitive information such as usernames, passwords, banking details, or personal data. The study focused on different phishing techniques, detection methods, and preventive measures.

4.2 SQL Injection (SQLi)

SQL Injection is a web application vulnerability that allows attackers to manipulate database queries by inserting malicious SQL commands into user input fields. Research included understanding vulnerable applications, attack methods, and secure coding practices to prevent SQL injection.

4.3 Cross-Site Scripting (XSS)

Cross-Site Scripting (XSS) is an attack where malicious JavaScript code is injected into a trusted website. This can lead to session hijacking, cookie theft, and unauthorized actions. The research covered Stored XSS, Reflected XSS, DOM-based XSS, and mitigation techniques.

4.4 Cross-Site Request Forgery (CSRF)

CSRF is an attack that forces an authenticated user to perform unwanted actions on a web application without their knowledge. The study focused on how CSRF attacks work, their impact, and protection mechanisms such as CSRF tokens.

4.5 Man-in-the-Middle (MITM) Attacks

A Man-in-the-Middle attack occurs when an attacker secretly intercepts communication between two parties to steal or modify transmitted data. Research included public Wi-Fi attacks, packet interception, SSL stripping, and encryption-based defenses.

4.6 Distributed Denial of Service (DDoS)

A DDoS attack overwhelms a server or network with a massive amount of traffic, making services unavailable to legitimate users. The research covered attack methods, botnets, traffic analysis, and mitigation strategies.

4.7 Password Security

Password security research focused on creating strong passwords, password entropy, password hashing, brute-force attacks, dictionary attacks, password managers, and multi-factor authentication (MFA) to improve account security.

4.8 Malware

Malware is malicious software designed to damage or gain unauthorized access to computer systems. Research included different types of malware such as viruses, worms, trojans, ransomware, spyware, adware, and rootkits, along with detection and prevention techniques.

4.9 MITRE ATT&CK Framework

The MITRE ATT&CK Framework is a globally recognized knowledge base of attacker tactics, techniques, and procedures (TTPs). It was studied to understand how cyber attacks are categorized, how security professionals analyze threats, and how organizations can improve their cyber defense strategies.

Summary

The research conducted during Month 1 provided a strong foundation for the CyberShield AI project by identifying major cyber security threats, understanding attack techniques, and selecting appropriate technologies and security practices that will be implemented in the upcoming development phases.

5. Technology Stack

The CyberShield AI project uses a modern and lightweight technology stack that is suitable for developing a secure, scalable, and easy-to-maintain web application. Each technology was selected based on its reliability, performance, and compatibility with cyber security applications.

5.1 Backend – Python 3.11 & Flask 3.x

The backend of the project is developed using **Python 3.11** and the **Flask 3.x** web framework. Python is widely used in cyber security because of its simplicity, extensive libraries, and strong community support. Flask is a lightweight framework that enables rapid development of REST APIs and web applications while providing flexibility for integrating security tools.

5.2 Database – SQLite & PostgreSQL

During the development phase, **SQLite** is used because it is lightweight, easy to configure, and requires no separate database server. For production deployment, **PostgreSQL** is selected as it provides better performance, scalability, reliability, and advanced security features for handling large amounts of data.

5.3 Object Relational Mapping (ORM) – SQLAlchemy

SQLAlchemy is used as the Object Relational Mapping (ORM) tool. It simplifies database operations by allowing developers to interact with the database using Python objects instead of writing raw SQL queries. It also improves code maintainability and helps reduce SQL-related errors.

5.4 Authentication – Flask-Login & JWT

The authentication system uses **Flask-Login** for secure user session management and **JSON Web Tokens (JWT)** for API authentication. This combination ensures secure login, role-based access control, and protected communication between users and the application.

5.5 Frontend – HTML, Jinja2, CSS & JavaScript

The user interface is built using **HTML**, **Jinja2 Templates**, **CSS**, and **JavaScript**. Jinja2 enables dynamic server-side rendering, while CSS and JavaScript provide an interactive, responsive, and user-friendly interface for accessing various cyber security tools.

5.6 Artificial Intelligence – Rule-Based Classifiers

The AI component is implemented using **rule-based classifiers**. Instead of complex machine learning models, predefined security rules and logical conditions are used to detect suspicious patterns, analyze user inputs, and provide explainable security recommendations. This approach is fast, lightweight, and does not require GPU resources.

5.7 Deployment – Docker & Docker Compose

The application is deployed using **Docker** and **Docker Compose**. Docker packages the application along with all required dependencies into containers, ensuring consistent performance across different environments. Docker Compose simplifies the deployment of multiple services such as the web application and database.

Summary

The selected technology stack provides a strong foundation for developing the CyberShield AI platform. It offers high performance, security, scalability, and ease of maintenance while supporting the implementation of cyber security tools, AI-powered analysis, and attack simulation features.

6. Modules

The **CyberShield AI** platform is divided into multiple functional modules, with each module designed to address a specific aspect of cyber security awareness, threat detection, and security training. These modules work together to provide a comprehensive learning and analysis platform for users.

6.1 Website / URL Scanner

The Website or URL Scanner allows users to check whether a website is safe before visiting it. The module analyzes URLs for potential phishing attempts, suspicious domains, insecure connections, and malicious links. It helps users identify potentially harmful websites and promotes safe browsing practices.

6.2 Email Phishing Analyzer

The Email Phishing Analyzer examines email content to identify phishing attempts. It analyzes sender information, suspicious links, email headers, and common phishing indicators. This module educates users about recognizing fraudulent emails and avoiding credential theft.

6.3 Password Strength Analyzer

The Password Strength Analyzer evaluates the security of user passwords by checking password length, complexity, character variety, and estimated resistance to brute-force attacks. It also provides suggestions for creating stronger and more secure passwords.

6.4 Malware File Scanner

The Malware Scanner allows users to upload files for security analysis. The module checks files for suspicious characteristics, known malware signatures, and potentially malicious behavior, helping users identify unsafe files before execution.

6.5 Network Scanner

The Network Scanner analyzes network devices, open ports, and available services within a network. It assists users in identifying potential security vulnerabilities and understanding the overall security status of their network environment.

6.6 Attack Simulator

The Attack Simulator provides a safe and controlled environment for demonstrating common cyber attacks without causing real-world damage. The simulator includes demonstrations of:

- SQL Injection (SQLi)
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- Man-in-the-Middle (MITM)
- Distributed Denial of Service (DDoS)
- Phishing Attacks

This module helps users understand how cyber attacks work and how they can be prevented.

6.7 Security Training and Quizzes

The Security Training module offers educational content, tutorials, and interactive quizzes covering various cyber security topics. It enables users to improve their knowledge, assess their understanding, and earn certificates upon successful completion.

6.8 AI Security Assistant (Chatbot)

The AI Security Assistant acts as an intelligent chatbot that answers cyber security-related questions, explains technical concepts, provides security recommendations, and guides users through different platform features.

6.9 Threat Intelligence Module

The Threat Intelligence module maintains information about known cyber threats, Indicators of Compromise (IOCs), and attack techniques based on frameworks such as **MITRE ATT&CK**. It helps users understand emerging threats and security trends.

6.10 Reports Module

The Reports module generates detailed security reports in multiple formats such as **PDF, Excel, and CSV**. These reports include scan results, detected vulnerabilities, user activity, and security recommendations for future reference.

6.11 Admin Panel

The Admin Panel provides administrative control over the entire CyberShield AI platform. Administrators can manage users, monitor system activity, review scan history, update training materials, manage threat intelligence data, and generate system-wide reports.

Summary

The modular architecture of CyberShield AI ensures that each feature is organized, scalable, and easy to maintain. Together, these modules provide users with a complete platform for cyber security awareness, threat detection, practical learning, and security management.

7. Database

The **CyberShield AI** platform uses a structured relational database to securely store user information, scan results, training records, and system activities. The database is designed to support authentication, cyber security analysis, threat intelligence, reporting, and user progress tracking. During development, **SQLite** is used for simplicity, while **PostgreSQL** is planned for production deployment due to its scalability and advanced security features.

7.1 Users Table

The **Users** table stores information about all registered users of the platform. It contains user details such as user ID, full name, email address, encrypted password, account status, registration date, and assigned role. This table is essential for user authentication and profile management.

7.2 Roles Table

The **Roles** table defines different user roles within the system, such as **Administrator, Security Analyst, and User**. It enables role-based access control (RBAC), ensuring that users can only access features permitted by their assigned role.

7.3 Scan History Table

The **Scan History** table stores records of all security scans performed by users. It includes information such as scan type, scan target, scan date and time, scan results, detected vulnerabilities, and recommendations. This allows users to review previous scan activities.

7.4 Audit Logs Table

The **Audit Logs** table records important user and system activities, including logins, logout events, password changes, administrative actions, and security-related operations. These logs help maintain accountability, improve monitoring, and support security investigations.

7.5 Threats Table

The **Threats** table maintains information about known cyber threats, attack techniques, threat categories, severity levels, and mitigation strategies. This data supports the Threat Intelligence module and helps users understand emerging cyber risks.

7.6 Indicators of Compromise (IOCs) Table

The **IOCs** table stores Indicators of Compromise such as malicious IP addresses, suspicious domains, file hashes, URLs, and email addresses. These indicators assist in identifying and detecting known cyber threats during security analysis.

7.7 Training Modules Table

The **Training Modules** table contains educational content related to cyber security awareness. It stores lesson titles, descriptions, learning materials, videos, and course categories, allowing users to access structured learning resources.

7.8 Quizzes Table

The **Quizzes** table stores quiz questions, multiple-choice options, correct answers, explanations, and associated training modules. This enables users to evaluate their knowledge after completing security training.

7.9 Certificates Table

The **Certificates** table stores information about certificates awarded to users after successfully completing training courses and quizzes. It includes certificate ID, user ID, course name, completion date, and certificate status.

7.10 Leaderboard Table

The **Leaderboard** table tracks user performance based on quiz scores, completed modules, and earned points. It encourages user engagement by displaying rankings and promoting healthy competition among learners.

Summary

The database design provides a secure and organized structure for managing user accounts, cyber security scans, learning resources, threat intelligence, reports, and system logs. It ensures efficient data management while supporting all major functionalities of the CyberShield AI platform.

8. Folder Structure

The **CyberShield AI** project follows a well-organized folder structure to ensure that the application is modular, scalable, and easy to maintain. Each folder and file has a specific purpose, allowing developers to manage different components of the application efficiently.

8.1 app.py

The **app.py** file is the main entry point of the application. It initializes the Flask application, loads the configuration settings, registers all routes (Blueprints), and starts the web server. It acts as the central controller of the project.

8.2 config.py

The **config.py** file stores all configuration settings required by the application, such as database connection details, secret keys, API keys, debugging options, and application environment variables. Keeping configuration separate improves security and maintainability.

8.3 database.py

The **database.py** file is responsible for configuring the database connection and initializing the database. It creates database sessions, manages database setup, and supports data seeding during development.

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8.4 models/

The **models** folder contains all database model classes created using SQLAlchemy. Each model represents a database table such as Users, Roles, Scan History, Threats, Training Modules, and other project entities. These models define the database structure and relationships.

8.5 routes/

The **routes** folder contains Flask Blueprints that define the application's URL endpoints and request handling logic. Each module, such as authentication, scanning, reporting, or training, has its own route file to keep the code modular and organized.

8.6 scanner/

The **scanner** folder contains the core scanning engines of the CyberShield AI platform. It includes modules for URL scanning, phishing detection, password analysis, malware scanning, and network scanning. These components perform the primary cyber security analysis functions.

8.7 ai/

The **ai** folder contains the artificial intelligence components used in the project. It includes rule-based classifiers, security analysis logic, and the AI Security Assistant (Chatbot), which provides intelligent recommendations and answers to users' cyber security questions.

8.8 templates/

The **templates** folder stores all HTML files used for the web interface. These templates are built using the Jinja2 templating engine, allowing dynamic content to be displayed based on user interactions and application data.

8.9 static/

The **static** folder contains all static resources used by the application, including CSS stylesheets, JavaScript files, images, icons, fonts, and other client-side assets required to build a responsive and attractive user interface.

8.10 tests/

The **tests** folder contains unit tests, integration tests, and security testing scripts. These tests help verify that all application features work correctly and ensure the reliability and stability of the system during development.

8.11 documentation/

The **documentation** folder stores all project-related documents, including project reports, seminar documents, technical documentation, API documentation, user manuals, database design, research papers, and development records. This folder helps maintain complete documentation throughout the project lifecycle.

Summary

The proposed folder structure provides a clean and modular architecture for the CyberShield AI project. It separates application logic, database models, scanning engines, AI components, templates, and documentation, making the project easier to develop, maintain, test, and scale in future versions.

9. References

The following references were studied during the research phase of the **CyberShield AI** project. These resources provide industry-standard guidelines, security frameworks, API documentation, and technical references that support the design and development of the platform.

9.1 OWASP Top 10

The **OWASP Top 10** is one of the most widely recognized cybersecurity standards for web application security. It identifies the ten most critical security risks affecting modern web applications, including SQL Injection, Cross-Site Scripting (XSS), Broken Authentication, Security Misconfiguration, and other common vulnerabilities. This reference was used to understand secure coding practices and web application security.

9.2 MITRE ATT&CK Framework

The **MITRE ATT&CK Framework** is a globally recognized knowledge base that documents attacker tactics, techniques, and procedures (TTPs) observed in real-world cyber attacks. It provides detailed information about how attackers operate and helps security professionals improve threat detection, incident response, and cyber defense strategies.

9.3 Have I Been Pwned (HIBP) API

The **Have I Been Pwned (HIBP) API** provides access to publicly known data breaches and compromised account information. It allows developers to check whether an email address or password has been exposed in previous data breaches. This API was studied to understand how breach detection can improve user security awareness.

9.4 Google Safe Browsing API

The **Google Safe Browsing API** enables applications to identify unsafe websites containing phishing attacks, malware, deceptive content, or harmful downloads. This reference was used for designing the Website/URL Scanner module to help users verify the safety of web links before visiting them.

9.5 VirusTotal API

The **VirusTotal API** is a cloud-based threat intelligence service that analyzes files, URLs, IP addresses, and domains using multiple antivirus engines and security vendors. It helps detect malware and other cyber threats by providing comprehensive security analysis. This API was studied for the Malware Scanner and URL analysis modules.

9.6 Flask Official Documentation

The **Flask Official Documentation** served as the primary reference for developing the web application. It provides detailed guidance on Flask architecture, routing, templates, request handling, authentication, database integration, REST APIs, and deployment. This documentation was used throughout the planning and development of the CyberShield AI platform.

Summary

These references provide a strong theoretical and practical foundation for the CyberShield AI project. They ensure that the platform follows industry best practices, incorporates recognized cybersecurity standards, and uses reliable technologies for secure web application development and cyber threat analysis.

10. Conclusion

The first month of the **CyberShield AI** project was successfully completed by establishing a strong foundation for the development of the platform. During this phase, the project title, objectives, and problem statement were finalized after identifying the growing need for cyber security awareness and threat detection solutions. Extensive research was conducted on major cyber security topics, including phishing attacks, SQL Injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), Man-in-the-Middle (MITM) attacks, Distributed Denial of Service (DDoS), password security, malware, and the MITRE ATT&CK Framework.

A suitable technology stack was selected, including **Python, Flask, SQLAlchemy, SQLite/PostgreSQL, HTML, CSS, JavaScript, Docker, and rule-based AI**, to ensure that the platform would be secure, scalable, and easy to maintain. The overall system architecture, project modules, database schema, and folder structure were carefully designed to support future development and expansion.

The successful completion of the planning and research phase provides a clear roadmap for the remaining stages of the project. With all major design decisions finalized, the project is now ready to move into the implementation phase.

In Month 2

the focus will shift toward practical development, including setting up the project environment, creating the Flask application, designing the database models, implementing user authentication (Register, Login, and JWT), and developing the core modules of the CyberShield AI platform. This structured approach will ensure systematic development and successful completion of the final-year project.